Project Report
On

Prediction Of Cardio Vascular Disease

Submitted By

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
S. R. K. R ENGINEERING COLLEGE BHIMAVARAM

CERTIFICATE

This is to certify that this is a bonafide work on "**PREDICTION OF CARDIOVASCULAR DISEASE**" as a **MEAN STACK TECHNOLOGIES AND DATA STRUCTURES** laboratory report submitted by Mr. D. N. V. DILEEP(21B91A0585), Mr. G. V. N. S. M. PRUDHVI RAJU(21B91A0594), Ms. G. LEELA SOWMYA(21B91A0595), Mr.I. P. B. PRAVEEN(21B91A05B5) in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering, during the academic year 2023-2024. The candidates worked right under our Supervision and guidance.

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Finally, yet importantly, we would like to express us heartful thanks to us beloved parents, faculty, us friends/classmates for their help, and our wishes for the successful completion of this project.

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TABLE OF CONTENTS

ABSTRACT

In our project "**Cardiovascular Disease Prediction**" we have tried to show the prediction of cardiovascular disease. It is the overview of prediction. It takes the information about age, gender, height, weight, bp, cholesterol, glucose level, smoking status, alcohol consumption, physical activity, bmi, pulse pressure and tells whether the person is suffering with the disease or not.This project is aimed to predict the cardiovascular disease of a person by first filling the personal credentials and the giving information about the credentials asked in the web.

Main objective of this project is to provide solution whether he/she is suffering from the disease. This software application will help the patients very well.The person had to signup first and can login with their username and password.This software has been made in a user friendly interface, so that anyone can singup And check whether they are suffering from cardiovascular disease. As a security we have provided Username and password.

MODULES

- Login page
- Sign up page
- Logout page
- Index page
- Connection PageApp page

SOFTWARE REQUIREMENTS

1. Text Editor : Visual Studio Code
2. Framework : Flask
3. Programming Languages : HTML,CSS,PHP
4. Internet Browser : Google Chrome/Microsoft Edge/Internet Browser

HARDWARE REQUIREMENTS

1. Processor-Intel Core I5
2. RAM-8GB

TECHNICAL STACK

HTML

HTML, or Hypertext Markup Language, is the foundational language used to create and design web pages. Introduced by Tim Berners-Lee in 1991, HTML enables the structuring of content on the internet, allowing text, images, videos, and other multimedia to be embedded and displayed in a web browser. HTML uses a system of tags and attributes to define elements and their properties. Each tag, enclosed in angle brackets, denotes a specific type of content or instruction, such as <p> for paragraphs, <a> for hyperlinks, and <img> for images. These elements are organized in a hierarchical structure, forming the Document Object Model (DOM), which browsers use to render web pages.

Over the years, HTML has evolved through several versions to incorporate new features and capabilities. HTML5, the latest major revision, introduced in 2014, brought significant enhancements, including support for audio and video elements, improved accessibility features, and more robust semantic tags like <article>, <section>, and <nav>. These updates allow developers to create richer, more interactive, and semantically meaningful web pages. HTML5 also emphasizes compatibility with mobile devices, ensuring responsive design and improved user experiences across different screen sizes and devices.

HTML works in conjunction with other web technologies such as CSS (Cascading Style Sheets) and JavaScript. While HTML provides the structure of a web page, CSS handles the presentation, defining styles, layouts, and visual effects. JavaScript, on the other hand, enables dynamic content and interactive functionality, such as form validation, animations, and real-time updates. Together, these technologies form the cornerstone of modern web development, empowering developers to create sophisticated and user-friendly websites and web applications.

CSS

CSS, or Cascading Style Sheets, is a stylesheet language used to control the presentation and layout of web pages. Introduced by the W3C in 1996, CSS allows developers to separate content from design by defining styles for HTML elements. This separation enhances maintainability and flexibility, enabling consistent styling across multiple pages and making it easier to update the look and feel of a website. CSS rules consist of selectors, which target HTML elements, and declarations, which specify the styles to be applied, such as color, font, and spacing.

One of the core advantages of CSS is its cascading nature, where multiple styles can be applied and prioritized based on specific rules. This cascade allows for styles to be inherited from parent elements and overridden by more specific selectors or inline styles when necessary. CSS also supports media queries, which enable responsive design by applying different styles based on the user's device characteristics, such as screen size and resolution. This functionality ensures that web pages are visually appealing and functional across a wide range of devices, from desktop computers to smartphones.

CSS has evolved significantly since its inception, with CSS3 introducing a host of new features and capabilities. These enhancements include advanced layout techniques like Flexbox and Grid, which offer more control over the arrangement of elements on a page. CSS3 also brought animations and transitions, allowing for smooth visual effects without relying on JavaScript. Additionally, CSS preprocessors like Sass and LESS have emerged, offering more powerful syntax and features like variables and nested rules, further streamlining the development process. Together, these advancements make CSS an essential tool for creating modern, responsive, and visually engaging web designs.

JAVA SCRIPT

JavaScript, often abbreviated as JS, is a dynamic programming language primarily used for adding interactivity and complex functionalities to web pages. Created by Brendan Eich in 1995 while working at Netscape Communications, JavaScript has evolved from a simple scripting language into one of the core technologies of the web, alongside HTML and CSS. Unlike HTML, which structures content, and CSS, which styles it, JavaScript enables developers to create interactive elements, manipulate the DOM, and handle events, thus bringing web pages to life. Common uses of JavaScript include form validation, dynamic content updates, and interactive graphics.

JavaScript's versatility extends beyond client-side scripting; it has become a powerful tool for server-side development through environments like Node.js. This allows developers to use JavaScript for building entire applications, from the server to the client, using a single language. JavaScript also supports various paradigms, including procedural, object-oriented, and functional programming, making it adaptable to a wide range of coding styles and applications. Its extensive ecosystem includes numerous frameworks and libraries, such as React, Angular, and Vue.js, which streamline and enhance the development process by providing reusable components and structure.

The evolution of JavaScript has been marked by the introduction of ECMAScript standards, with ECMAScript 6 (ES6) being a significant milestone. ES6 introduced features like arrow functions, classes, modules, and promises, which simplify code and enhance functionality. Additionally, modern JavaScript tools and workflows, including package managers like npm, build tools like Webpack, and transpilers like Babel, have further empowered developers to create sophisticated and efficient.

PHP

PHP, initially developed by Rasmus Lerdorf in 1994, stands for "Hypertext Preprocessor." It's a server-side scripting language primarily designed for web development, but it can also be used as a general-purpose programming language. PHP scripts are executed on the server, generating HTML content that is then sent to the client's browser, enabling dynamic and interactive web experiences. PHP is widely supported across various operating systems and web servers, making it one of the most popular choices for building dynamic websites and web applications.

One of PHP's key strengths is its versatility and ease of use. It integrates seamlessly with HTML, allowing developers to embed PHP code directly into HTML pages for dynamic content generation. PHP supports a wide range of databases, including MySQL, PostgreSQL, and SQLite, enabling developers to interact with databases to store and retrieve information dynamically. Additionally, PHP offers extensive libraries and frameworks, such as Laravel, Symfony, and CodeIgniter, which streamline development and provide solutions for common tasks like authentication, routing, and templating.

PHP has undergone significant improvements over the years, with each new version introducing features aimed at improving performance, security, and developer productivity. PHP 7, released in 2015, brought substantial performance enhancements, making PHP applications significantly faster and more efficient. The latest versions continue to build upon this foundation, with features like scalar type declarations, return type declarations, and the null coalescing operator, further enhancing the language's capabilities and expressiveness.

SAMPLE CODES

1.login.php

```php
<?php

session_start();

include("connection.php");
include("functions.php");

if($_SERVER['REQUEST_METHOD'] == "POST")
{
   //something was posted
   $user_name = $_POST['user_name'];
   $password = $_POST['password'];

   if(!empty($user_name) && !empty($password) && !is_numeric($user_name))
   {
      //read from database
      $query = "select * from users where user_name = '$user_name' limit 1";
      $result = mysqli_query($con, $query);

      if($result)
      {
         if($result && mysqli_num_rows($result) > 0)
         {
            $user_data = mysqli_fetch_assoc($result);

            if($user_data['password'] === $password)
            {
               $_SESSION['user_id'] = $user_data['user_id'];
               header("Location: index.html");
               die;
            }
         }
      }

      echo "Wrong username or password!";
   }
   else
   {
      echo "Wrong username or password!";
   }
}
```

```
?>

<!DOCTYPE html>
<html>
<head>
  <title>Login</title>
  <style type="text/css">
    #container {
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
    }

    #box {
      background-color: grey;
      width: 300px;
      padding: 20px;
      border-radius: 15px;
    }

    .input-group {
      margin-bottom: 15px;
    }

    label {
      display: block;
      margin-bottom: 5px;
      color: white;
    }

    #text {
      height: 25px;
      border-radius: 5px;
      padding: 4px;
      border: solid thin #aaa;
      width: 100%;
    }

    #button, #signup-button {
      padding: 10px;
      width: 100%;
      color: white;
      background-color: lightblue;
      border: none;
      border-radius: 5px;
      cursor: pointer;
    }

    #signup-button {
      background-color: blue; /* Changed color to blue */
      margin-top: 10px; /* Added margin to separate from the Login button */
```

```
        }
    </style>
</head>
<body>

    <div id="container">
        <div id="box">
            <form method="post">
                <div style="font-size: 20px; margin-bottom: 20px; color: white;">Login</div>

                <div class="input-group">
                    <label for="user_name">Username:</label>
                    <input id="text" type="text" name="user_name" required>
                </div>

                <div class="input-group">
                    <label for="password">Password:</label>
                    <input id="text" type="password" name="password" required>
                </div>

                <input id="button" type="submit" value="Login">
                <a href="signup.php"><input id="signup-button" type="button" value="Signup"></a>
            </form>
        </div>
    </div>

</body>
</html>
```

2.Signup.php

```
<?php
session_start();

	include("connection.php");
	include("functions.php");


	if($_SERVER['REQUEST_METHOD'] == "POST")
	{
		//something was posted
		$user_name = $_POST['user_name'];
		$password = $_POST['password'];

		if(!empty($user_name) && !empty($password) && !is_numeric($user_name))
		{

			//save to database
			$user_id = random_num(20);
```

```
                $query = "insert into users (user_id,user_name,password) values
('$user_id','$user_name','$password')";

                mysqli_query($con, $query);

                header("Location: login.php");
                die;
            }else
            {
                echo "Please enter some valid information!";
            }
        }
?>


<!DOCTYPE html>
<html>
<head>
    <title>Signup</title>
</head>
<body>

    <style type="text/css">

    #text{

        height: 25px;
        border-radius: 5px;
        padding: 4px;
        border: solid thin #aaa;
        width: 100%;
    }

    #button{

        padding: 10px;
        width: 100px;
        color: white;
        background-color: lightblue;
        border: none;
    }

    #box{

        background-color: grey;
        margin: auto;
        width: 300px;
        padding: 20px;
    }

    </style>
```

```
	<div id="box">

		<form method="post">
			<div style="font-size: 20px;margin: 10px;color: white;">Signup</div>

			<input id="text" type="text" name="user_name"><br><br>
			<input id="text" type="password" name="password"><br><br>

			<input id="button" type="submit" value="Signup"><br><br>

			<a href="login.php">Click to Login</a><br><br>
		</form>
	</div>
</body>
</html>
```

3.logout.php

```
<?php

session_start();

if(isset($_SESSION['user_id']))
{
	unset($_SESSION['user_id']);

}

header("Location: login.php");
die;
```

4. index.html

```
<!DOCTYPE html>
<html>
  <head>
    <title>My website</title>
    <style>
      #container {
        width: 50%;
        margin: auto;
        border-radius: 15px;
        padding: 20px;
        background-color: #f0f0f0;
      }
```

```
    .input-group {
      margin-bottom: 15px;
      border-radius: 15px;
      padding: 10px;
      background-color: #e0e0e0;
    }

    label {
      display: block;
      margin-bottom: 5px;
      font-weight: bold;
    }

    h2 {
      font-weight: bold;
    }

    #buttons {
      text-align: center;
    }

    #predict-button,
    #logout-button {
      padding: 10px 20px;
      margin: 0 10px;
      border: none;
      border-radius: 5px;
      cursor: pointer;
    }

    #predict-button {
      background-color: lightblue;
    }

    #logout-button {
      background-color: #ff6347;
    }

    #output-box {
      margin-top: 20px;
      padding: 15px;
      background-color: #e0e0e0;
      border-radius: 10px;
    }
  </style>
</head>
<body>
  <div id="container">
    <br />
    <p>Hello</p>

    <!-- Form for feature input -->
```

```html
<form onsubmit="handleSubmit(event)">
  <div class="input-group">
    <label for="age">Age (in years):</label>
    <input type="number" name="age" required />
  </div>

  <div class="input-group">
    <label for="gender">Gender:</label>
    <select name="gender" required>
      <option value="1">Male</option>
      <option value="0">Female</option>
    </select>
  </div>

  <div class="input-group">
    <label for="height">Height (cm):</label>
    <input type="number" name="height" required />
  </div>

  <div class="input-group">
    <label for="weight">Weight (kg):</label>
    <input type="number" name="weight" required />
  </div>

  <div class="input-group">
    <label for="ap_hi">Systolic blood pressure (ap_hi):</label>
    <input type="number" name="ap_hi" required />
  </div>

  <div class="input-group">
    <label for="ap_lo">Diastolic blood pressure (ap_lo):</label>
    <input type="number" name="ap_lo" required />
  </div>

  <div class="input-group">
    <label for="cholestrol">Cholesterol level:</label>
    <select name="cholestrol" required>
      <option value="1">Normal</option>
      <option value="2">Above Normal</option>
      <option value="3">Well Above Normal</option>
    </select>
  </div>

  <div class="input-group">
    <label for="gluc">Glucose level:</label>
    <select name="gluc" required>
      <option value="1">Normal</option>
      <option value="2">Above Normal</option>
      <option value="3">Well Above Normal</option>
    </select>
  </div>
```

```html
<div class="input-group">
  <label for="smoke">Smoke:</label>
  <select name="smoke" required>
    <option value="0">No</option>
    <option value="1">Yes</option>
  </select>
</div>

<div class="input-group">
  <label for="alco">Alcohol consumption:</label>
  <select name="alco" required>
    <option value="0">No</option>
    <option value="1">Yes</option>
  </select>
</div>

<div class="input-group">
  <label for="active">Physical activity:</label>
  <select name="active" required>
    <option value="0">No</option>
    <option value="1">Yes</option>
  </select>
</div>

<div class="input-group">
  <label for="bmi">BMI:</label>
  <input type="number" name="bmi" step="0.01" required />
</div>

<div class="input-group">
  <label for="bmi_category">BMI Category:</label>
  <select name="bmi_category" required>
    <option value="1">Underweight</option>
    <option value="2">Normal</option>
    <option value="3">Overweight</option>
    <option value="4">Obese</option>
  </select>
</div>

<div class="input-group">
  <label for="pulse_pressure">Pulse Pressure:</label>
  <input type="number" name="pulse_pressure" required />
</div>

<!-- <div class="input-group">
    <label for="map">mean absolute pressure:</label>
    <input type="number" name="map" required />
  </div> -->

<div id="buttons">
  <input type="submit" value="Predict" id="predict-button" />
  <a href="logout.php" id="logout-button">Logout</a>
```

```
        </div>
      </form>
      <div id="output-box">
        <h2>Prediction Result:</h2>
        <p style="font-size: 24px"><?php echo $prediction_result; ?></p>
      </div>
    </div>
  </body>
  <script>
    function handleSubmit(event) {
      event.preventDefault();
      const form = event.target;
      const formData = new FormData(form);
      const data = {};
      for (let [key, value] of formData.entries()) {
        data[key] = value;
      }
      console.log(data);
      fetch("http://127.0.0.1:5000/predict", {
        method: "POST",
        body: JSON.stringify(data),
        headers: {
          "Content-Type": "application/json",
        },
      })
        .then((response) => response.json())
        .then((data) => {
          document.getElementById("output-box").innerHTML = `
            <h2>Prediction Result:</h2>
            <p style="font-size: 24px">${data.message}</p>
          `;
        });
    }
  </script>
</html>
```

5.connection.php

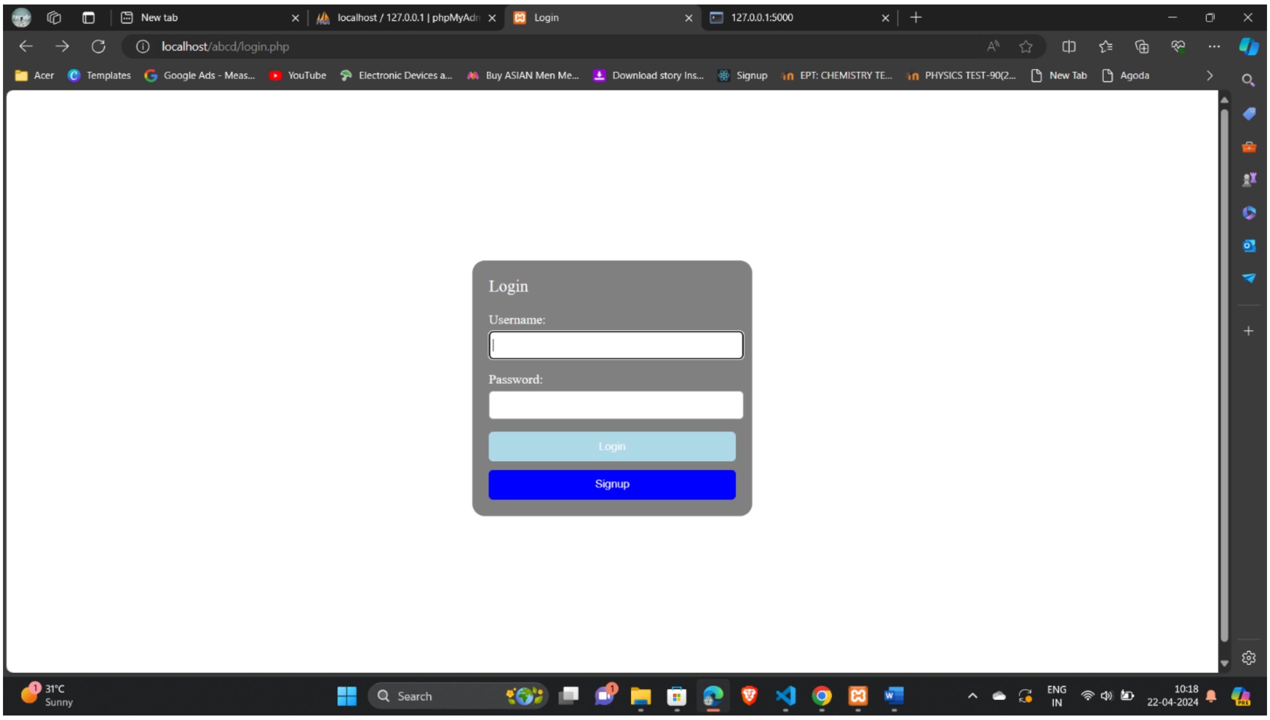
```
<?php

$dbhost = "localhost";
$dbuser = "root";
$dbpass = "";
$dbname = "login_sample_db";

if(!$con = mysqli_connect($dbhost,$dbuser,$dbpass,$dbname))
{

	die("failed to connect!");
```

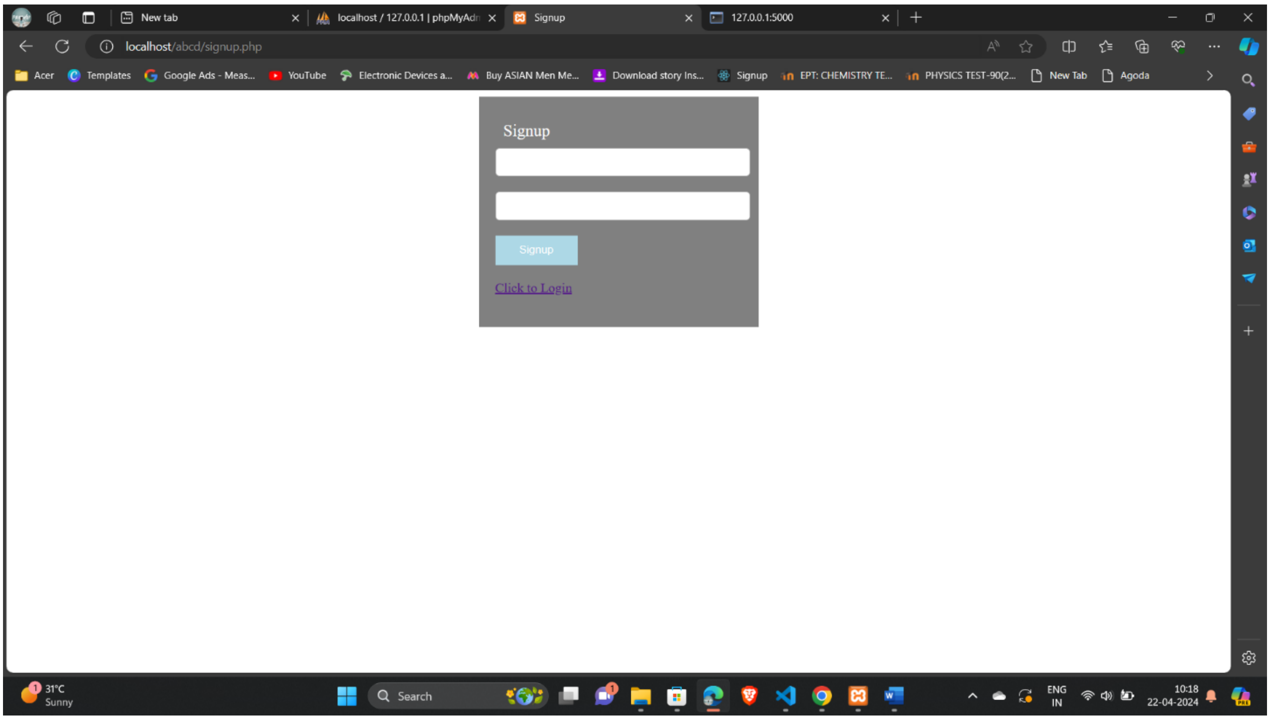
}

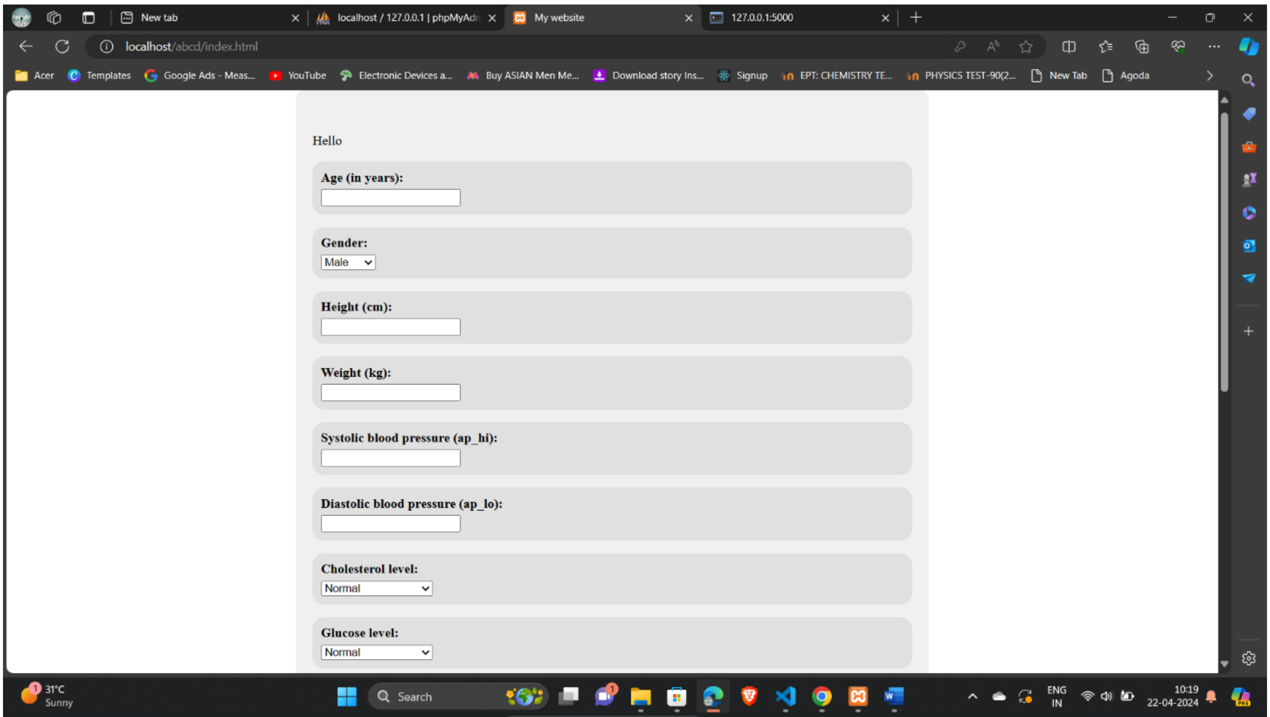
6.app.py

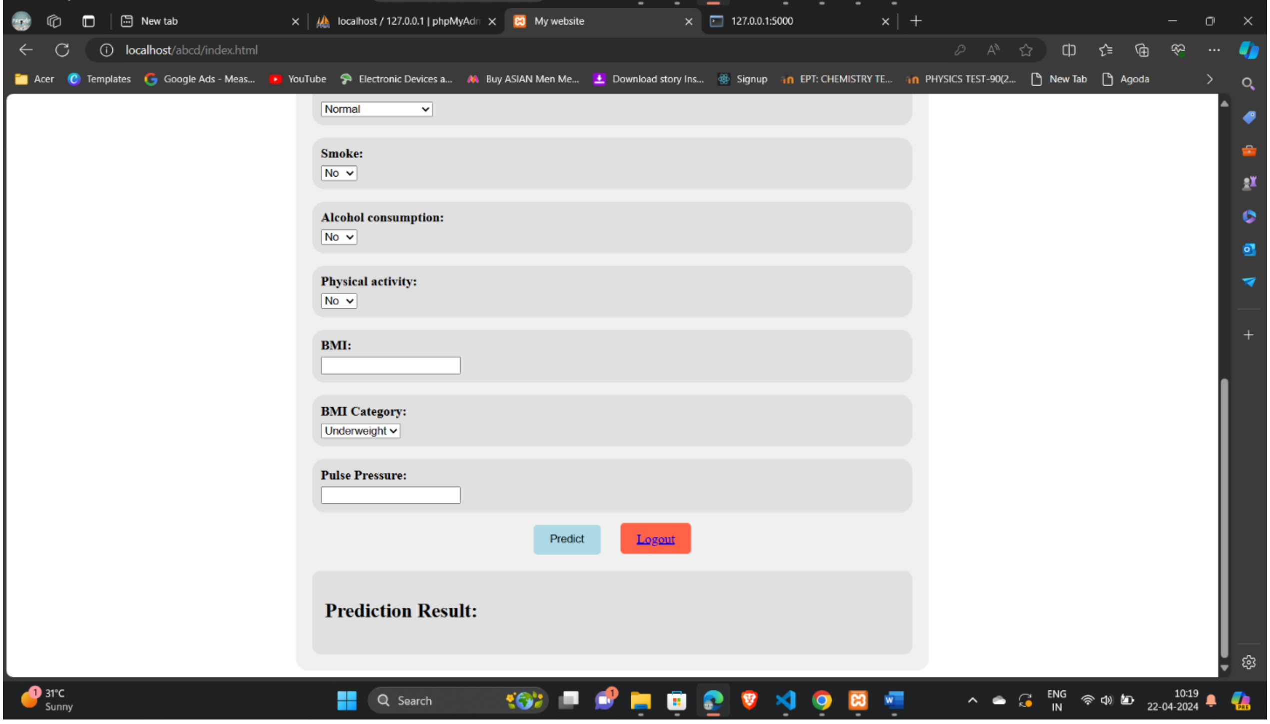
```
from flask import Flask
import joblib
from flask import request
from flask_cors import CORS
from flask import jsonify

app = Flask(__name__)
CORS(app)

model = joblib.load('best_model_pelican.pkl')

@app.route('/')
def hello():
    return 'Hello, World!'

@app.route("/predict", methods=['POST'])
def predict():
    # Get data from request
    data = request.get_json()
    print(data)

    map=(float(data['ap_lo']) + (float(data['ap_hi']) -float(data['ap_lo'])) / 3)
    # Use the data for prediction
    X = [[float(data['age']), float(data['gender']), float(data['height']), float(data['weight']),
float(data['ap_hi']),
          float(data['ap_lo']), float(data['cholestrol']), float(data['gluc']), float(data['smoke']),
float(data['alco']),
          float(data['active']), float(data['bmi']), float(data['bmi_category']),
float(data['pulse_pressure']), map]]
    y_pred = model.predict(X)
    print(y_pred)
    # Return the prediction result
    if y_pred[0] == 0:
        return jsonify({"message": "Predicted class: No Cardiovascular disease"})
    else:
        return jsonify({"message": "Predicted class: Cardiovascular disease"})

if __name__ == '__main__':
    app.run(debug=True)
```

SCREENSHOTS

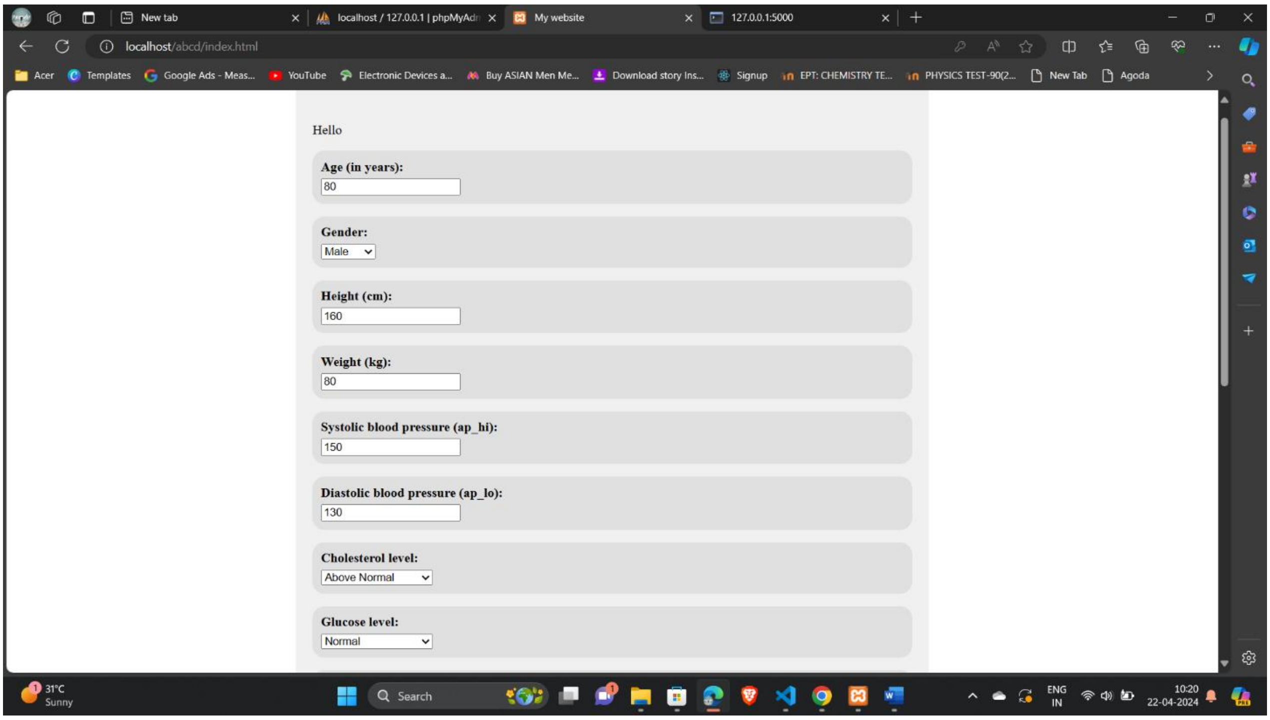


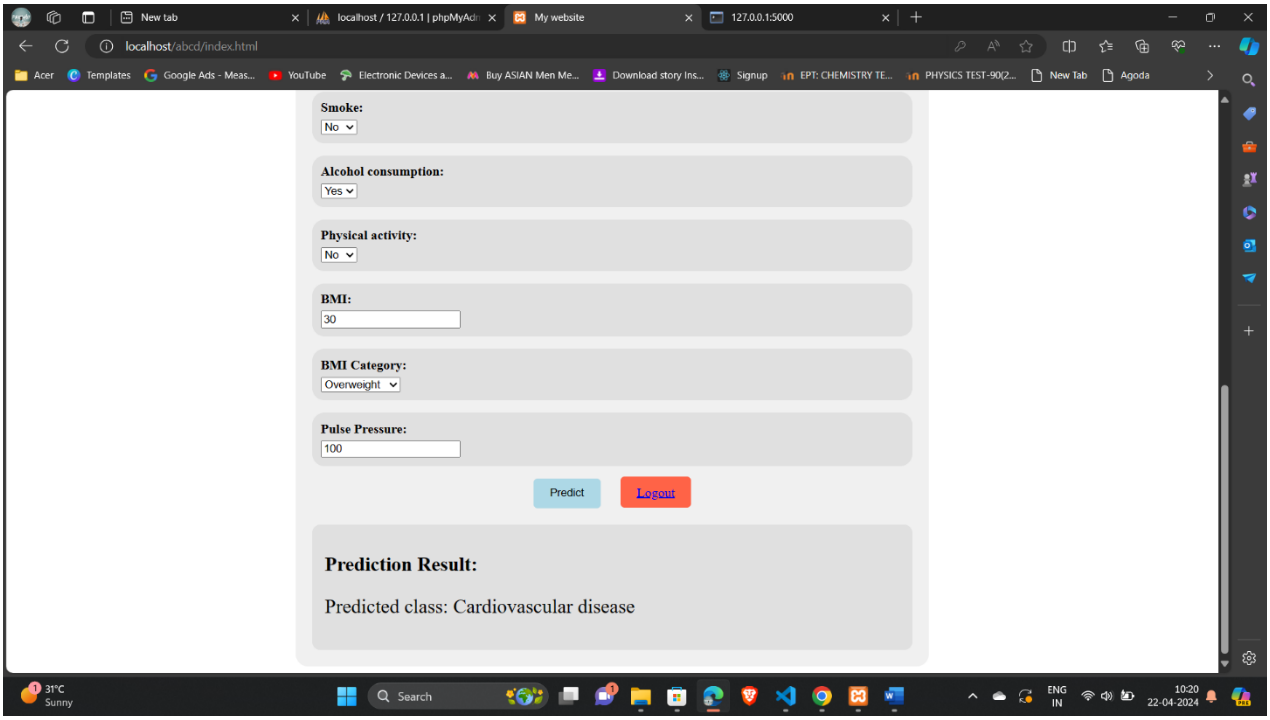



localhost/abcd/index.html
Hello
Age (in years):
Gender:
Male
Height (cm):
Weight (kg):
Systolic blood pressure (ap_hi):
Diastolic blood pressure (ap_lo):
Cholesterol level:
Normal
Glucose level:
Normal


localhost/abcd/index.html
Normal
Smoke:
No
Alcohol consumption:
No
Physical activity:
No
BMI:
BMI Category:
Underweight
Pulse Pressure:
Predict
Logout
Prediction Result:

SAMPLE OUTPUT





CONCLUSION

In this project we have tried our best to make user friendly software. This software can be handled by any person who has little bit of idea of computers. In this software we have tried to meet most of the requirements to know about the disease according to present generation. Maintaining details of username nd password.

In our effort we have tried to make our software all the more user friendly but there may some features which we would like to include in our continuous attempts.